In [27]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

In [28]:
```
df = pd.read_csv("C:\\Users\\Admin\\Desktop\\E-commerce Customer Behavior.csv")
```

In [29]:
```
df
```

Out[29]:

| | Customer ID | Gender | Age | City | Membership Type | City Sales | Items Purchased | Average Rating | Dis A |
|---|---|---|---|---|---|---|---|---|---|
| **0** | 101 | Female | 29 | New York | Gold | 1120.20 | 14 | 4.6 | |
| **1** | 102 | Male | 34 | Los Angeles | Silver | 780.50 | 11 | 4.1 | |
| **2** | 103 | Female | 43 | Chicago | Bronze | 510.75 | 9 | 3.4 | |
| **3** | 104 | Male | 30 | San Francisco | Gold | 1480.30 | 19 | 4.7 | |
| **4** | 105 | Male | 27 | Miami | Silver | 720.40 | 13 | 4.0 | |
| **...** | ... | ... | ... | ... | ... | ... | ... | ... | |
| **345** | 446 | Male | 32 | Miami | Silver | 660.30 | 10 | 3.8 | |
| **346** | 447 | Female | 36 | Houston | Bronze | 470.50 | 8 | 3.0 | |
| **347** | 448 | Female | 30 | New York | Gold | 1190.80 | 16 | 4.5 | |
| **348** | 449 | Male | 34 | Los Angeles | Silver | 780.20 | 11 | 4.2 | |
| **349** | 450 | Female | 43 | Chicago | Bronze | 515.75 | 10 | 3.3 | |

350 rows × 11 columns

In [31]:
```
df.describe()
```

Out[31]:

| | Customer ID | Age | City Sales | Items Purchased | Average Rating | Days Since Last Purchase |
|---|---|---|---|---|---|---|
| **count** | 350.000000 | 350.000000 | 350.000000 | 350.000000 | 350.000000 | 350.000000 |
| **mean** | 275.500000 | 33.597143 | 845.381714 | 12.600000 | 4.019143 | 26.588571 |
| **std** | 101.180532 | 4.870882 | 362.058695 | 4.155984 | 0.580539 | 13.440813 |
| **min** | 101.000000 | 26.000000 | 410.800000 | 7.000000 | 3.000000 | 9.000000 |
| **25%** | 188.250000 | 30.000000 | 502.000000 | 9.000000 | 3.500000 | 15.000000 |
| **50%** | 275.500000 | 32.500000 | 775.200000 | 12.000000 | 4.100000 | 23.000000 |
| **75%** | 362.750000 | 37.000000 | 1160.600000 | 15.000000 | 4.500000 | 38.000000 |
| **max** | 450.000000 | 43.000000 | 1520.100000 | 21.000000 | 4.900000 | 63.000000 |

In [32]:
```
df.isnull().sum()
```

Out[32]:
```
Customer ID                0
Gender                     0
Age                        0
City                       0
Membership Type            0
City Sales                 0
Items Purchased            0
Average Rating             0
Discount Applied           0
Days Since Last Purchase   0
Satisfaction Level         2
dtype: int64
```

In [33]:
```
df['Age'].fillna(df['Age'].median(),inplace=True)
```

```
C:\Users\Admin\AppData\Local\Temp\ipykernel_16188\1527141296.py:1: FutureWarning:
A value is trying to be set on a copy of a DataFrame or Series through chained as
signment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work becau
se the intermediate object on which we are setting values always behaves as a cop
y.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.meth
od({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to pe
rform the operation inplace on the original object.


  df['Age'].fillna(df['Age'].median(),inplace=True)
```

In [34]:
```
df
```

Out[34]:

| | Customer ID | Gender | Age | City | Membership Type | City Sales | Items Purchased | Average Rating | Dis A |
|---|---|---|---|---|---|---|---|---|---|
| **0** | 101 | Female | 29 | New York | Gold | 1120.20 | 14 | 4.6 | |
| **1** | 102 | Male | 34 | Los Angeles | Silver | 780.50 | 11 | 4.1 | |
| **2** | 103 | Female | 43 | Chicago | Bronze | 510.75 | 9 | 3.4 | |
| **3** | 104 | Male | 30 | San Francisco | Gold | 1480.30 | 19 | 4.7 | |
| **4** | 105 | Male | 27 | Miami | Silver | 720.40 | 13 | 4.0 | |
| **...** | ... | ... | ... | ... | ... | ... | ... | ... | |
| **345** | 446 | Male | 32 | Miami | Silver | 660.30 | 10 | 3.8 | |
| **346** | 447 | Female | 36 | Houston | Bronze | 470.50 | 8 | 3.0 | |
| **347** | 448 | Female | 30 | New York | Gold | 1190.80 | 16 | 4.5 | |
| **348** | 449 | Male | 34 | Los Angeles | Silver | 780.20 | 11 | 4.2 | |
| **349** | 450 | Female | 43 | Chicago | Bronze | 515.75 | 10 | 3.3 | |

350 rows × 11 columns

In [35]:
```
df.drop_duplicates(inplace=True)
```

In [36]:
```
df
```

Out[36]:

| | Customer ID | Gender | Age | City | Membership Type | City Sales | Items Purchased | Average Rating | Dis A |
|---|---|---|---|---|---|---|---|---|---|
| **0** | 101 | Female | 29 | New York | Gold | 1120.20 | 14 | 4.6 | |
| **1** | 102 | Male | 34 | Los Angeles | Silver | 780.50 | 11 | 4.1 | |
| **2** | 103 | Female | 43 | Chicago | Bronze | 510.75 | 9 | 3.4 | |
| **3** | 104 | Male | 30 | San Francisco | Gold | 1480.30 | 19 | 4.7 | |
| **4** | 105 | Male | 27 | Miami | Silver | 720.40 | 13 | 4.0 | |
| **...** | ... | ... | ... | ... | ... | ... | ... | ... | |
| **345** | 446 | Male | 32 | Miami | Silver | 660.30 | 10 | 3.8 | |
| **346** | 447 | Female | 36 | Houston | Bronze | 470.50 | 8 | 3.0 | |
| **347** | 448 | Female | 30 | New York | Gold | 1190.80 | 16 | 4.5 | |
| **348** | 449 | Male | 34 | Los Angeles | Silver | 780.20 | 11 | 4.2 | |
| **349** | 450 | Female | 43 | Chicago | Bronze | 515.75 | 10 | 3.3 | |

350 rows × 11 columns

In [37]:
```
df = df[df['City Sales'] > 0]
```

In [38]:
```
df
```

Out[38]:

| | Customer ID | Gender | Age | City | Membership Type | City Sales | Items Purchased | Average Rating | Di A |
|---|---|---|---|---|---|---|---|---|---|
| **0** | 101 | Female | 29 | New York | Gold | 1120.20 | 14 | 4.6 | |
| **1** | 102 | Male | 34 | Los Angeles | Silver | 780.50 | 11 | 4.1 | |
| **2** | 103 | Female | 43 | Chicago | Bronze | 510.75 | 9 | 3.4 | |
| **3** | 104 | Male | 30 | San Francisco | Gold | 1480.30 | 19 | 4.7 | |
| **4** | 105 | Male | 27 | Miami | Silver | 720.40 | 13 | 4.0 | |
| **...** | ... | ... | ... | ... | ... | ... | ... | ... | |
| **345** | 446 | Male | 32 | Miami | Silver | 660.30 | 10 | 3.8 | |
| **346** | 447 | Female | 36 | Houston | Bronze | 470.50 | 8 | 3.0 | |
| **347** | 448 | Female | 30 | New York | Gold | 1190.80 | 16 | 4.5 | |
| **348** | 449 | Male | 34 | Los Angeles | Silver | 780.20 | 11 | 4.2 | |
| **349** | 450 | Female | 43 | Chicago | Bronze | 515.75 | 10 | 3.3 | |

350 rows × 11 columns

In [39]:
```
City_Sales = df.groupby('City')['Membership Type'].sum()
```

In [40]:
```
print(City_Sales)
```

```
City
Chicago          BronzeBronzeBronzeBronzeBronzeBronzeBronzeBron...
Houston          BronzeBronzeBronzeBronzeBronzeBronzeBronzeBron...
Los Angeles      SilverSilverSilverSilverSilverSilverSilverSilv...
Miami            SilverSilverSilverSilverSilverSilverSilverSilv...
New York         GoldGoldGoldGoldGoldGoldGoldGoldGoldGoldGoldGo...
San Francisco    GoldGoldGoldGoldGoldGoldGoldGoldGoldGoldGoldGo...
Name: Membership Type, dtype: object
```

In [41]:
```
total_Sales = df['City Sales'].sum()
print(total_Sales)
```

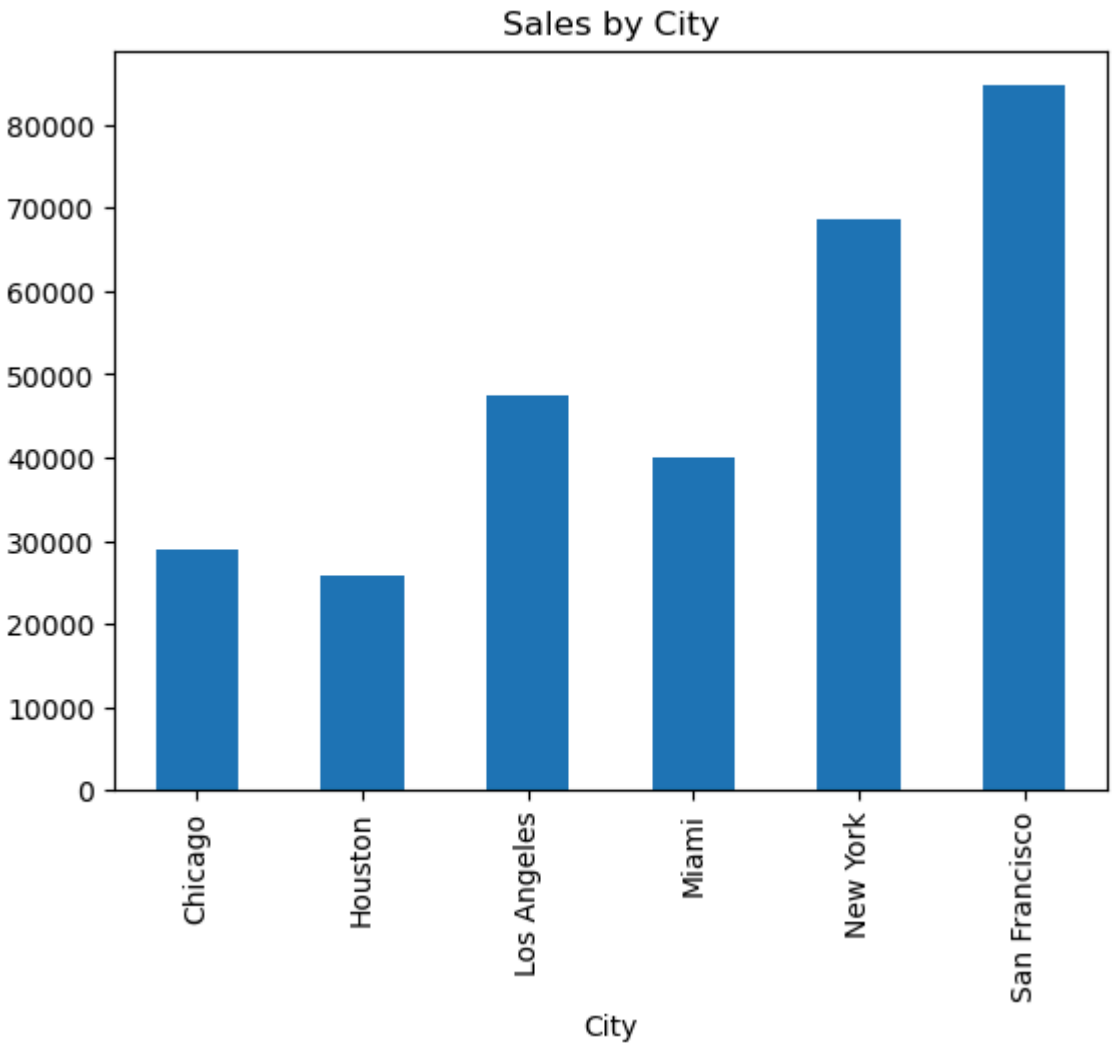
```
295883.6
```

In [42]:
```
Gender_Avg = df.groupby('Gender')['City Sales'].sum()
print(Gender_Avg)
```

```
Gender
Female    123170.0
Male      172713.6
Name: City Sales, dtype: float64
```

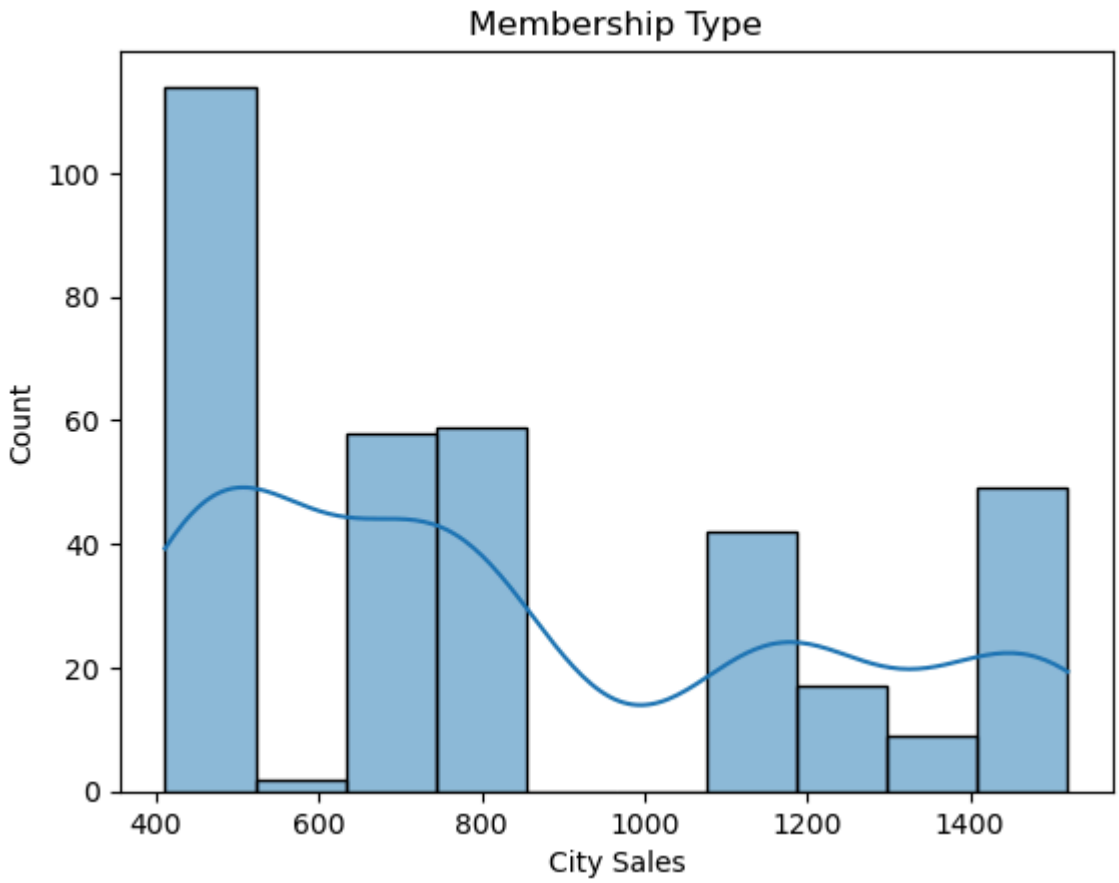
In [45]:
```
City_Sales = df.groupby('City')['City Sales'].sum()
City_Sales.plot(kind='bar')
plt.title("Sales by City")
plt.show
```

Out[45]: <function matplotlib.pyplot.show(close=None, block=None)>



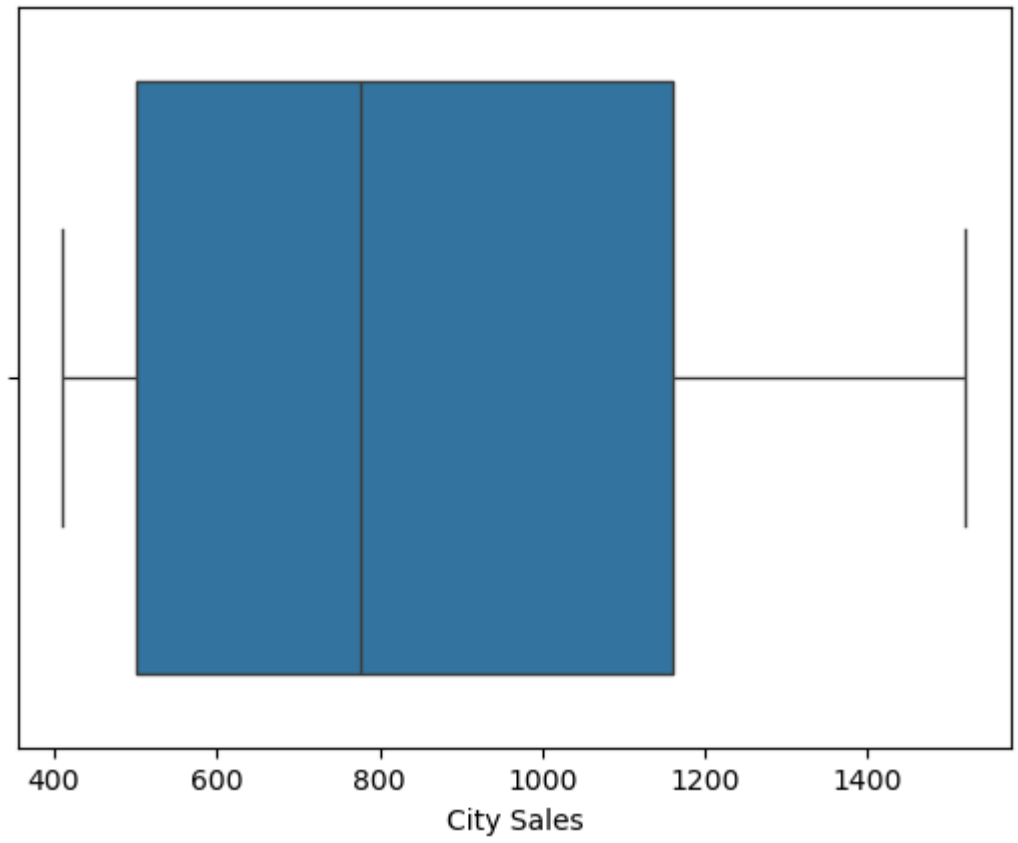

In [46]:
```
sns.histplot(df['City Sales'], kde=True)
plt.title("Membership Type")
plt.show
```

Out[46]: <function matplotlib.pyplot.show(close=None, block=None)>



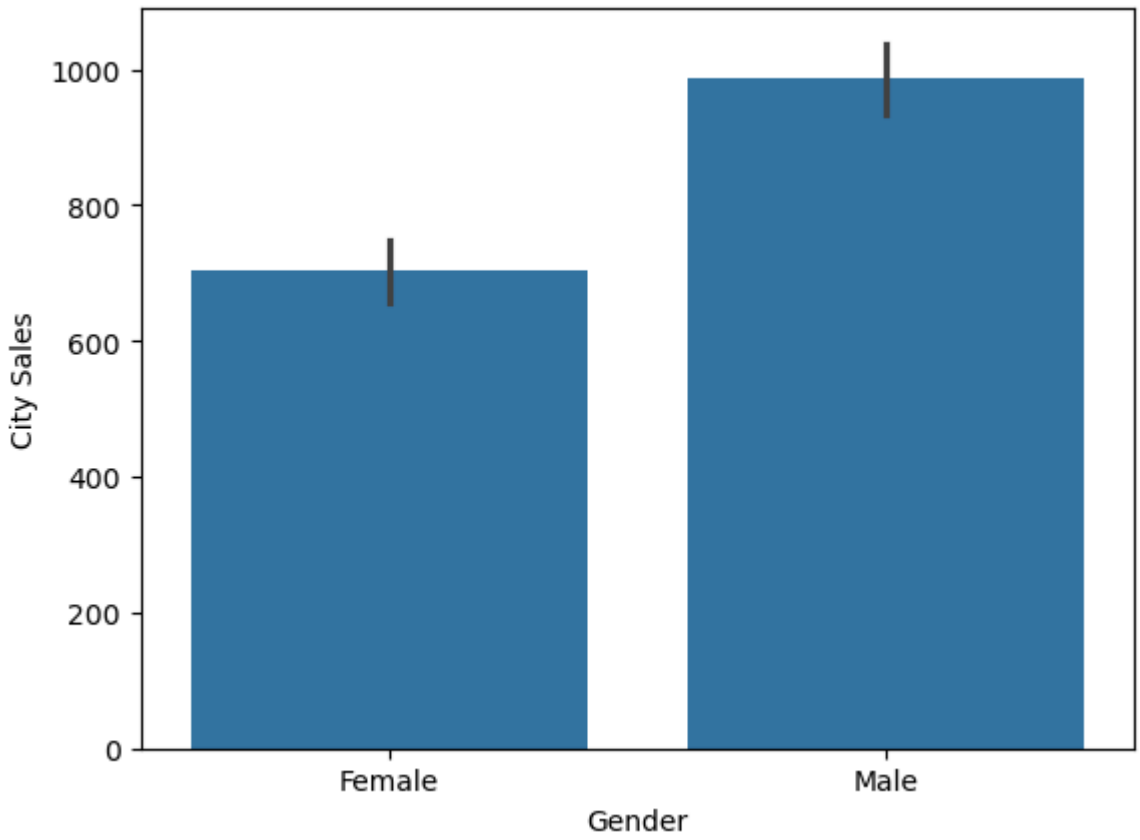

In [47]:
```
sns.boxplot(x=df['City Sales'])
plt.show
```

Out[47]:
```
<function matplotlib.pyplot.show(close=None, block=None)>
```



In [49]:
```
sns.barplot(x='Gender', y='City Sales' ,data=df)
plt.show
```

Out[49]:
```
<function matplotlib.pyplot.show(close=None, block=None)>
```




In [ ]: